

SOUTHEAST SHOAL, ON, Canada

WMO: 714660

Lat: 41.8267N

Lon: 82.4632W

Elev: 195

StdP: 99.00

Time Zone: -5.00 (NAE)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-12.1	-20.6	0.6	-13.2	-18.0	0.8	-10.5	21.2	-6.2	19.4	-6.7	11.6	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	27.9	23.7	26.8	23.0	25.7	21.9	24.9	26.6	23.8	25.3	22.8	24.6	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	19.8	26.0	23.3	18.6	25.0	22.2	17.3	24.1	76.9	26.6	72.4	25.3	68.4	24.7	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	16.0	14.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-2.5	-1.3	1.4	7.0	13.8	19.5	22.3	22.1	18.5	12.8	6.1	0.4
	DBStd	9.86	5.56	5.62	4.76	3.74	4.10	3.58	2.88	2.44	3.55	3.80	4.15	4.94
	HDD10.0	1533	387	316	271	104	10	0	0	0	1	18	128	298
	HDD18.3	3387	644	549	526	341	152	29	4	2	41	176	367	555
	CDD10.0	1561	1	1	3	13	127	284	382	377	256	103	11	2
	CDD18.3	372	0	0	0	0	11	64	128	121	46	3	0	0
	CDH23.3	1173	0	0	0	0	17	162	539	391	63	1	0	0
	CDH26.7	110	0	0	0	0	1	19	63	23	3	0	0	0
Wind	WSAvg	7.2	8.9	7.5	7.6	7.4	6.5	5.7	5.7	5.6	6.7	7.9	8.5	8.7
Precipitation	PrecAvg	956	64	55	67	92	94	100	95	88	90	70	72	71
	PrecMax	1162	142	135	138	137	183	159	218	215	206	119	139	184
	PrecMin	675	19	14	28	51	31	34	41	20	31	19	21	31
	PrecStd	125	31	30	27	25	43	35	40	45	40	24	26	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	13.4	16.3	18.7	25.2	28.9	30.0	28.5	26.6	22.1	16.2	13.3
		MCWB	10.8	9.0	10.5	12.8	18.6	22.7	25.0	23.6	21.8	18.0	12.4	12.4
	2%	DB	8.9	10.2	12.9	15.9	23.2	26.7	28.2	27.3	24.8	20.5	14.5	11.0
		MCWB	8.0	8.7	9.4	11.3	17.8	21.9	24.5	22.9	20.6	17.0	11.2	9.4
	5%	DB	7.0	8.0	9.8	13.8	21.5	25.1	27.2	26.3	24.0	19.1	13.1	8.6
		MCWB	6.2	6.4	6.8	9.9	16.9	21.4	24.1	22.3	20.3	15.8	10.2	6.6
	10%	DB	4.4	5.3	7.3	12.0	19.6	23.8	26.0	25.4	23.1	17.6	11.8	6.6
		MCWB	3.2	3.2	4.6	8.6	15.9	20.7	22.6	21.6	19.4	14.3	9.1	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	9.6	12.3	14.2	20.2	24.5	26.4	26.0	23.6	19.4	13.9	12.3
		MCDB	11.3	11.4	15.1	17.0	22.9	26.0	28.0	27.0	24.3	20.7	15.5	13.0
	2%	WB	7.7	6.2	10.0	12.2	18.7	23.2	25.2	24.7	22.4	17.9	12.2	8.7
		MCDB	8.3	7.7	12.6	14.0	22.2	24.9	27.0	26.0	23.6	19.8	13.9	9.6
	5%	WB	5.2	3.6	7.5	10.5	17.4	22.1	24.2	23.6	21.2	16.3	10.5	5.7
		MCDB	5.7	4.6	9.9	12.9	20.3	24.0	25.9	25.1	23.0	18.3	12.8	6.8
	10%	WB	2.2	2.3	5.2	9.1	16.0	21.1	23.0	22.6	20.2	14.9	9.2	4.0
		MCDB	3.0	3.2	7.3	11.7	19.1	23.1	24.8	24.6	22.5	17.0	11.4	5.4
Mean Daily Temperature Range	5% DB	MDBR	4.9	4.5	5.1	5.3	5.1	4.4	4.0	3.8	4.1	4.4	4.1	3.9
		MCDBR	6.8	6.2	9.3	8.5	7.4	5.6	4.8	4.2	4.3	5.4	5.7	5.0
		MCWBR	7.6	7.5	7.1	5.8	4.5	4.0	4.4	4.3	4.2	4.7	5.4	5.3
	5% WB	MCDBR	7.8	6.2	9.5	8.1	6.8	5.0	4.5	4.1	3.9	5.7	5.9	5.0
		MCWBR	7.8	6.2	7.8	6.4	5.0	4.3	4.8	4.9	4.7	5.4	5.6	5.3
Clear-Sky Solar Irradiance	taub	0.307	0.332	0.356	0.397	0.430	0.444	0.438	0.424	0.382	0.366	0.325	0.314	
	taud	2.391	2.288	2.303	2.267	2.218	2.228	2.266	2.293	2.388	2.407	2.486	2.445	
	Ebn at Noon	837	865	887	872	848	835	836	837	851	820	814	798	
	Edh at Noon	81	104	116	130	140	139	132	125	106	93	74	71	
All-Sky Solar Radiation	RadAvg	1.38	2.21	3.42	4.60	5.43	6.08	6.09	5.33	4.29	2.78	1.73	1.18	
	RadStd	0.15	0.28	0.31	0.36	0.45	0.39	0.33	0.21	0.37	0.22	0.25	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	N/A

Nomenclature: See separate page